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REPORT

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Hybrid AI/ML-mechanistic framework enables intelligent optimization of commercial biopharmaceutical downstream processing

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ABSTRACT

Biopharmaceutical manufacturing requires continuous improvement to ensure robust, efficient, and high-quality processes, yet traditional experimental designs remain resource-demanding and insufficient to capture interactions of multiple parameters. Here, we introduce a hybrid framework integrating artificial intelligence (AI)/machine learning (ML) with mechanistic modeling to optimize anion-exchange chromatography and resolve the long-standing yield-purity trade-off in PEGylated protein purification. Three critical process parameters were first identified through correlation analysis between 30 input factors and critical quality attributes/process yield from 400+ commercial manufacturing lots, which were further refined using equilibrium dispersive and steric mass action models. Over 40,000 *in silico* optimization via the mechanistic model resolved the yield-purity trade-off, achieving a 12% increase in yield and 33% reduction in high-molecular-weight impurities. The optimized process conditions were verified across laboratory ($n = 3$), pilot ($n = 3$), and commercial ($n = 18$) runs, consistently demonstrating scalability and process robustness. This study highlights the power of combining data-driven machine learning with mechanistic modeling for process optimization, leading to an improved commercial process with substantial cost savings and paving the way for upcoming intelligent biomanufacturing.

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Introduction

Biopharmaceutical manufacturing underpins modern healthcare by delivering life-saving therapies ranging from recombinant proteins to advanced gene and cell therapies. As biologics become increasingly complex and global demand continues to grow, process optimization has emerged as a critical challenge for the industry. Continuous improvement is not only a regulatory expectation, as emphasized by ICH Q10, but also a strategic imperative that unites patient needs, healthcare sustainability, and industrial competitiveness through enhanced product quality and reduced costs.

Conventional optimization approaches, including one-factor-at-a-time experiments and design of experiments (DoE), face inherent limitations in this context.[1] These methods are resource-intensive (often requiring more than 6 months to complete) and statistically constrained by dimensionality reduction, limiting their ability to capture complex multivariate interactions. As a result, they frequently fail to achieve the efficiency and precision demanded by modern biologics production, underscoring the urgent need for more advanced, data-driven strategies with well-understood mechanisms to guide next-generation manufacturing.

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Consequently, the biopharmaceutical industry has increasingly turned to artificial intelligence/machine learning (AI/ML) to overcome these limitations. Driven by the need for higher efficiency, reduced variability, and improved product quality, AI/ML applications in biopharmaceutical manufacturing have expanded rapidly in recent years.[2–11] Algorithms such as multivariate data analysis and artificial neural networks (ANNs) have been used to analyze large-scale process data, predict outcomes, and optimize operating conditions in real time.[6,7,11] For example, ANN-based models have accurately predicted depth filter loading capacity for monoclonal antibody clarification ($R^2 = 0.98$), enabling variable filter sizing and achieving a 10% reduction in cost of goods.[2] Similarly, ANN-driven optimization of CHO cell culture has led to a 48% increase in antibody production.[10] Beyond upstream and clarification steps, recent studies have also explored the potential of integrating AI with Internet of things (IoT) systems to enhance efficiency, quality control, and regulatory compliance. Collectively, these advances highlight the transformative promise of AI/ML in biopharmaceutical processing, while underscoring the need for approaches that combine predictive power with mechanistic interpretability to achieve reliable, scalable impact.[12]

In contrast, mechanistic models offer interpretable descriptions of the underlying transport and binding phenomena and have been successfully applied to optimize ion-exchange[13–19] hydrophobic interaction,[20,21] mixed-mode chromatography,[22–24] and continuous chromatography.[25–28] These models capture convection, dispersion, liquid – solid mass transfer, pore diffusion, and adsorption, requiring relatively little experimental data for calibration and enabling extrapolation beyond measured conditions. Recent advances have even enabled digital twin systems for chromatographic control, where mechanistic models accurately predicted elution profiles and optimized collection windows in real time, maintaining product quality with <2% error.[18] However, when applied to highly complex processes with numerous variables, mechanistic models can become computationally intensive and difficult to manage, which constrains their broader industrial application.

This dichotomy highlights a critical gap: the lack of a unified framework that combines the pattern-recognition strength of AI/ML with the interpretability and extrapolation capabilities of mechanistic modeling.[29–31] To address this need, we propose a hybrid workflow that strategically integrates AI/ML with mechanistic modeling to optimize the PEGylated protein purification via anion-exchange chromatography (AEX). Unlike previous approaches used in isolation, our methodology (Figure 1) establishes a synergistic loop: AI/ML rapidly screens extensive manufacturing datasets to identify critical process parameters, while mechanistic modeling provides interpretable, mechanism-based simulations for *in silico* optimization and sensitivity analysis. This hybrid design bridges the "black-box" nature of AI/ML with the physical insight of mechanistic modeling, while simultaneously reducing the computational burden of comprehensive mechanistic modeling.

The objectives of this study are four-fold: (1) develop a robust parameter-screening protocol combining correlation analysis with ML-based evaluation; (2) establish a mechanistic model to predict chromatographic profiles across diverse operating conditions; (3) implement a stepwise optimization strategy balancing performance with associated risk; and (4) verify the optimized process across laboratory, pilot, and commercial scales. This work would make three key contributions: it systematically leverages underutilized commercial manufacturing data for process optimization; it establishes a general framework for integrating data-driven and mechanistic approaches; and it provides a regulatory-compatible pathway for implementing process improvements through multi-scale verification. By breaking the traditional yield-purity trade-off and maintaining stringent quality control (<1% high molecular weight (HMW) level), this framework delivers measurable industrial benefits while ensuring compliance with regulatory expectations. Importantly, it demonstrates how concepts often confined to laboratory studies can be successfully transferred to commercial-scale manufacturing, paving the way for intelligent, lifecycle-oriented bioprocess intensification.

The workflow begins with systematic data collection from manufacturing datasets, followed by rigorous data review and preprocessing to ensure accuracy and reliability. ML tools are applied to identify the critical and key process parameters (CPPs and KPPs), which are subsequently optimized using mechanistic modeling. Then the optimization outcomes are confirmed in scale-down studies and further confirmed at pilot and commercial manufacturing scales. The workflow operates as a continuous cycle of data collection,

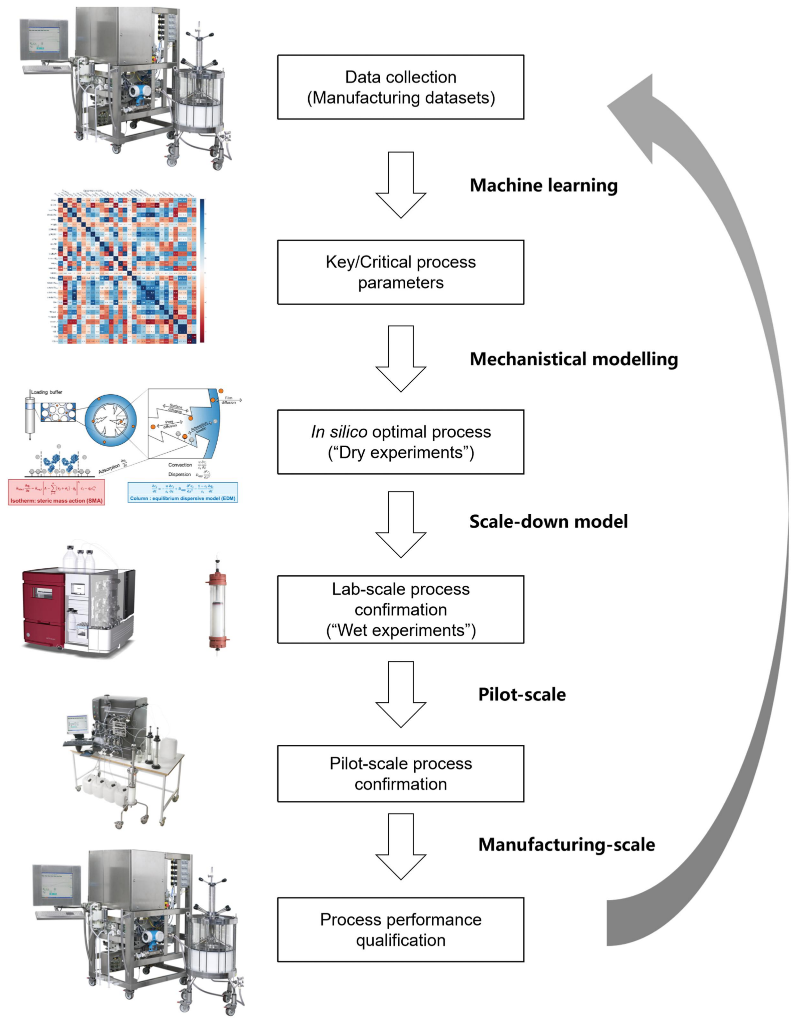

Figure 1. Hybrid workflow integrating AI/ML and mechanistic modeling for process optimization. The workflow begins with systematic data collection from manufacturing datasets, followed by rigorous data review and preprocessing to ensure accuracy and reliability. ML tools are applied to identify the critical and key process parameters (CPPs and KPPs), which are subsequently optimized using mechanistic modeling. Then the optimization outcomes are confirmed in scale-down studies and further confirmed at pilot and commercial manufacturing scales. The workflow operates as a continuous cycle of data collection, analysis, optimization, and verification, enabling iterative process improvement and robust lifecycle management.

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Materials and methods

Machine learning

Objective of the ML workflow

The primary objective of this research is to harness ML for enhancing the efficiency of biopharmaceutical downstream processes by accurately predicting critical quality attributes (CQAs) and key process attributes, such as purity and yield, from various process input parameters. Building on the identification of critical parameters, we then used strategies including Latin hypercube sampling (LHS), grid sampling, Bayesian optimization, and genetic algorithms to generate extensive parameter combinations within their physical and engineering constraints. These combinations are subjected to high-throughput prediction and screening using a pre-trained ML model to identify optimal parameter ranges that align with desired outcomes. The selected parameter sets are subsequently verified through lab-scale experiments and, upon demonstrating stability and reliability, are applied to pilot or commercial production. This integrated approach enables a model-driven, closed-loop process optimization, where ML algorithms dynamically regulate process inputs to ensure desired outcomes, minimize variability, and enhance overall product quality.

Data collection

The data collection for this study was designed to construct models that accurately reflect real-world manufacturing conditions. To achieve this, a substantial dataset was assembled, comprising 423 individual production batches monitored over a 13-month period from August 2023 to August 2024. This extended timeframe ensured that natural process variations and diverse operational conditions were comprehensively captured. Each batch in the dataset was characterized by 30 distinct process parameters, which are fundamental levers for controlling and optimizing process outcomes. These parameters were systematically classified into different categories based on their function and influence. Direct input parameters are those actively adjusted to steer the process toward its targets. In contrast, indirect input parameters represent background factors that, while not directly controlled, can still significantly affect system variability and performance. The resulting process output parameters are used to evaluate the success and results of each production run. Among these outputs, a specific CQA, namely HMW species, is of paramount importance. This CQA is related to the safety, efficacy, and quality of the product, making it the central focus of any process optimization effort. This ensures both regulatory compliance and product reliability, particularly when efforts are made to improve process yield.

Data preparation

Data preparation constituted a critical phase in establishing a reliable foundation for model development. The process began with rigorous data cleaning, where batches containing missing quality attributes or incomplete process parameter records were systematically excluded, refining the initial dataset to a finalized set of 400 representative batches. Subsequently, all parameters were logically categorized into distinct groups – including direct inputs, indirect inputs, outputs, and CQAs – to support structured analysis and model interpretation. To mitigate the potential distortion introduced by parameters of varying scales and units, normalization and scaling techniques were applied, ensuring that no single parameter disproportionately influenced model behavior, and thereby enhancing both the stability and predictive accuracy of the resulting models.

Model building workflow

The ML workflow was systematically designed to address the research objective of predicting CQAs from process parameters. This comprehensive process began with clearly defining the predictive modeling task, followed by exploratory data analysis to uncover underlying patterns and relationships within the dataset. Feature engineering was then used to refine and select the most informative predictors, enhancing the model's ability to capture essential signals in the data. For model development, multiple algorithms were rigorously evaluated including random forest[32] for handling complex interactions, support vector machines[33] for high-dimensional spaces, linear regression for baseline performance, and advanced gradient boosting methods like XGBoost,[34] and LightGBM[35] for their efficiency with large datasets. Neural networks[36] were also considered for capturing highly non-linear relationships. The final phase involved robust validation through train-test splitting and k-fold (setting $k = 5$ here) cross-validation,[37] ensuring reliable performance estimation and minimizing overfitting risks across diverse data partitions.

In this procedure, the total dataset was randomly partitioned into five equal-sized subsamples (folds). The training and validation process was then executed as follows: (1) Iterative Training: The model was trained five separate times. In each iteration, four of the folds (80% of the data) were used as the training set, while the remaining single fold (20% of the data) was retained as the validation set. (2) Rotational Validation: This process was rotated so that each of the five folds served as the validation set exactly once. (3) Aggregation: The performance results from all five folds were averaged to produce a single estimation of the model's generalization performance.

Model evaluation

Model evaluation constituted a critical phase in validating the reliability of ML models for predicting CQAs. A comprehensive set of performance metrics was used to thoroughly assess model capabilities, including accuracy for overall prediction correctness, precision and recall for evaluating positive class identification, and the F1 score to balance these measures, particularly in imbalanced datasets. For regression tasks, R-squared quantified the proportion of explained variance, while mean squared error and its rooted counterpart measured prediction deviations. The analysis further incorporated Pearson correlation coefficient to assess linear relationships, Spearman correlation coefficient for rank-based monotonic associations, and mean absolute percentage error for intuitive interpretation of prediction errors. Through systematic comparison across these diverse metrics, the relative strengths and limitations of different algorithmic approaches were rigorously examined. Based on the overall rankings of the Pearson correlation coefficient, Spearman correlation coefficient, and mean absolute percentage error, the most appropriate model for CQA prediction in the biopharmaceutical manufacturing context of this study was determined to the best of our knowledge.

Model optimization

Model optimization was conducted through a comprehensive approach to enhance performance and reliability. The process involved fine-tuning hyperparameters and feature selection (e.g., ablation study based on CQA correlation strength) to identify best configurations that optimize performance metrics. Additionally, various algorithms, including random forest, support vector machines, linear regression, XGBoost, LightGBM, and neural networks, were tested (as claimed above in the Model Building Workflow section), and ensemble methods were used to leverage the strengths of multiple models, thereby improving overall performance. For instance, we have summarized the comparative performance of the candidate models for yield prediction in the Supplementary Table S1. This table provides a head-to-head comparison of the various model architectures and features. This quantitative analysis identified Model Candidates 3332 and 3401 as the superior architecture and feature combinations.

To ensure robustness and stability, five-fold cross-validation techniques were utilized, enabling thorough evaluation of the model's consistency across different datasets. This iterative refinement process not only improved predictive accuracy but also ensured that the models could perform effectively under diverse conditions. By systematically analyzing performance metrics and incorporating advanced optimization techniques, the models achieved superior reliability and precision in their predictions.

Model deployment

Visualization techniques were strategically employed to elucidate the relationships between predicted and actual values of critical process parameters. This analytical approach encompassed the use of correlation heatmaps to illustrate both linear and monotonic associations among parameters and CQAs through Pearson and Spearman coefficients. Complementing this, SHapley Additive exPlanations (SHAP) value visualizations systematically quantified the contribution of individual features to model predictions, thereby enhancing both interpretability and transparency in the modeling outcomes.[38]

By integrating the results from correlation heatmaps and the model's feature importance, we identified critical process parameters essential for generating input parameter combinations. These combinations aim to deploy the best-performing model for optimal parameter search. Based on the selected critical parameters and their associated physical or engineering constraints, extensive parameter combinations are generated using techniques such as LHS, grid sampling, Bayesian optimization, or genetic algorithms. Over 10 billion parameter combinations were created and subjected to high-throughput prediction and virtual screening using the pre-trained optimal model to identify those aligning with desired CQAs.

Resin, protein, and buffers

The AEX steps were performed using NW Rose Q BB resin (NanoMicro, Suzhou, China) packed in columns of three distinct scales: laboratory (XK column, 1.6 cm × 23.4 cm), pilot (Easy-Axi column, 10.0 cm × 23.4 cm), and manufacturing scale (BPG column, 45 cm × 23.4 cm). To ensure process consistency and minimize

Table 1. Calibration and verification experiments for mechanistic modeling.

No.	Purpose	Loading density (g/L)	Elution length (CV)	Elution starting point	Elution ending point
Run 1	Calibration	1.0	15	0%B	60%B
Run 2	Calibration	1.0	20	0%B	60%B
Run 3	Calibration	2.5	15	0%B	60%B
Run 4	Calibration	3.8	15	0%B	60%B
Run 5	Verification	3.8	15	35%B	35%B

scale-up risks, the protein intermediate and stock solutions were sourced directly from in-house manufacturing operations. All required buffers for the chromatography process, including equilibration, wash, and elution solutions, were subsequently prepared according to established formulations using these stock solutions.

Calibration and verification experiments for mechanistic modeling

The AEX experiments were conducted on an ÄKTA avant 150™ chromatography system (Cytiva, Uppsala, Sweden). For mechanistic modeling, the buffer A contained 20 mM Tris-HCl (pH 7.4), and the buffer B consisted of 20 mM Tris-HCl and 130 mM NaCl (pH 7.4). All of the experiments consisted of equilibration (3 column volumes (CV) of buffer A), sample loading, wash (3 CV of buffer A), elution (0-60% buffer B) and regeneration (2 CV of 1 M NaCl) with a linear velocity of 150 cm/h. A detailed summary of the loading density and elution step of each experiment employed for model calibration and verification is presented in Table 1. During the elution steps, the protein peaks were collected in each 0.5 CV fraction that was further analyzed. The total protein concentration in each fraction was quantified using an ultraviolet spectrophotometer (UV1900i, Shimadzu Corporation, Japan). The size exclusion chromatography – high performance liquid chromatography (SEC-HPLC) analytical method was used to obtain the content of HMW and mono-PEGylated protein in each fraction.

Mechanistic model calibration and verification

The mechanistic model inside the AEX column combined the equilibrium dispersive model (EDM) for mass transfer and the steric mass action (SMA) model for adsorption. The detailed descriptions of these models and the determination of their parameters (ε_t, D_{app}, and Λ) are provided in our previous publication.[15] Based on the four calibration experiments, an inverse fitting method was used to estimate the SMA model parameters (ν, k_{eq}, k_{kin}, and σ) for HMW and mono-PEGylated protein. For model verification, one stepwise elution experiment (see Table 1, Run 5) was used to assess the predictive capability of the mechanistic model beyond the calibration range. All mechanistic model parameters in this study are presented in Table 2.

Process confirmation and robustness evaluation

When the optimization target was achieved by *in silico* simulation, the process was confirmed by "wet" experiments at both laboratory and pilot scales. Both scales were carried out using the identical procedure,

Table 2. Mechanistic model parameters.

Parameter	Notation	Value	Unit
Column length	L	234	mm
Linear velocity	u	0.42	mm/s
Ionic capacity	Λ	1.61	mol/L
Total porosity	ε_t	0.85	–
Apparent dispersion coefficient	D_{app}	0.12	mm^2/s
Characteristic charge	ν	$1.02/1.45^{\dagger}$	–
Equilibrium coefficient	k_{eq}	0.11/0.08	–
Kinetic coefficient	k_{kin}	3.52/0.76	sM^{ν}
Shielding factor	σ	$2.39 \times 10^4/1.52 \times 10^4$	–

†1.02/1.45 means that the characteristic charge of HMW and mono-PEGylated protein were 1.02 and 1.45, respectively.

as described here. The chromatographic process was performed with equilibration using 20 mM Tris-HCl, pH 7.4. The sample was conditioned with 20 mM Tris-HCl and 4 M NaCl to adjust protein concentration and conductivity prior to loading. After sample application, the column was washed with 20 mM Tris-HCl, 4 mM NaCl, pH 7.4, followed by elution with 20 mM Tris-HCl, 50 mM NaCl, pH 7.4. The eluate was analyzed by UV absorbance for protein concentration and by SEC-HPLC for HMW level.

To test the robustness of the process, wash conductivity and wash column volume were evaluated by a full-factorial DoE design. The lower and upper limits of wash conductivity were set as 1.8 mS/cm and 2.2 mS/cm, and those of the wash column volume set as 4 CV and 8 CV. Triplicate runs were performed at the set point, thus seven runs in total for the DoE design. Among all the experiments, the pH and conductivity of load kept the same with those of wash. The results were analyzed by JMP, using process yield and HMW level as the outputs.

Process performance qualification

Six consecutive tests for process performance qualification (PPQ) were carried out to verify that product quality and process performance could be consistently controlled. The PPQ protocol was drafted based on internal guidelines and aligned with the National Medical Products Administration's Technical Guidelines for Post-Approval Changes to Marketed Biological Products. This work only discusses the AEX step and focuses on the process yield and HMW level of the AEX to compare the results of the pre-change and post-change processes.

SEC-HPLC

In this study, the detection of HMW level was performed using HPLC according to a standardized operating procedure. The HPLC system (Agilent Technologies, CA), equipped with a TSKgel-G3000 SW_{XL} column (Tosoh Bioscience, Japan), was operated under the following conditions: a mobile phase of 0.5% triethylamine − 0.05 mol/L phosphate buffer (pH 7.0), a detection wavelength of 214 nm, a flow rate of 0.6 mL/min, and a column temperature of ambient temperature. Samples were injected at a volume of 20 µL and analyzed over a 30-min run time. Sample solutions were prepared by diluting the test product in the mobile phase to a concentration of approximately 1.0 mg/mL. System suitability tests were conducted to ensure the validity of the chromatographic conditions, and data were analyzed using area normalization method for HMW level.

Results

Hybrid workflow for synergistic optimization of a manufacturing process

We developed a hybrid workflow that integrates ML and mechanistic modeling for the synergistic optimization of biopharmaceutical manufacturing process (Figure 1). The strategy is designed to achieve measurable process improvements with minimal changes, guided by *in silico* optimization and subsequently verified through PPQ runs. By combining data-driven parameter identification with physics-based simulation, the workflow provides a robust path for process optimization that is in alignment with regulatory expectations.

This integrated workflow begins with AI/ML-driven parameter identification, where a systematic analysis of manufacturing data establishes correlations between process parameters and CQAs or yield, effectively screening the most influential parameters for subsequent optimization. The second phase involves mechanistic modeling and *in silico* optimization, employing models calibrated through experiments based on mass transfer and adsorption principles. Guided by the ML-identified parameters, a progressive optimization strategy is implemented: starting with adjustments within approved ranges, then extending to parameters with minimal impact on CQAs, and ultimately expanding to additional parameters after appropriate regulatory engagement. The third stage combines computational simulations with necessary experimental verification to assess process robustness, establishing reliable operating windows through wet-lab confirmation since current regulatory frameworks do not fully accept unvalidated models. The workflow concludes with process verification and regulatory alignment, where laboratory and pilot-scale confirmation runs support the execution of process verification and comparability studies to demonstrate robustness and secure regulatory approval.

Key/Critical process parameter determination by AI/ML

A robust ML framework was established to predict CQAs, such as HMW species and impurities. Host cell protein and host cell DNA are no longer considered CQAs at this purification step, as both were consistently below the quantification limit in the protein solution prior to the PEGylation process. Therefore, this study focused specifically on optimizing yield and HMW removal. The development process was underpinned by a systematic methodology that included exploratory data analysis, correlation analysis, feature engineering, and rigorous model evaluation. To elucidate the interdependencies between process parameters and CQAs, both Pearson and Spearman correlation matrices were constructed (Figure 2a,b). Leveraging these correlation heatmaps, this systematic methodology provided a clear, data-driven basis for selecting the most impactful input features, a critical step that substantially enhanced the final model's predictive power.

The selected features (shown in Table 3) include process parameters (PPs) such as conductivity, pH, and UV indicators, as well as column performance indicators and sample characteristics. The following features demonstrated their importance in achieving optimal model performance. The heterogeneity of the PEG

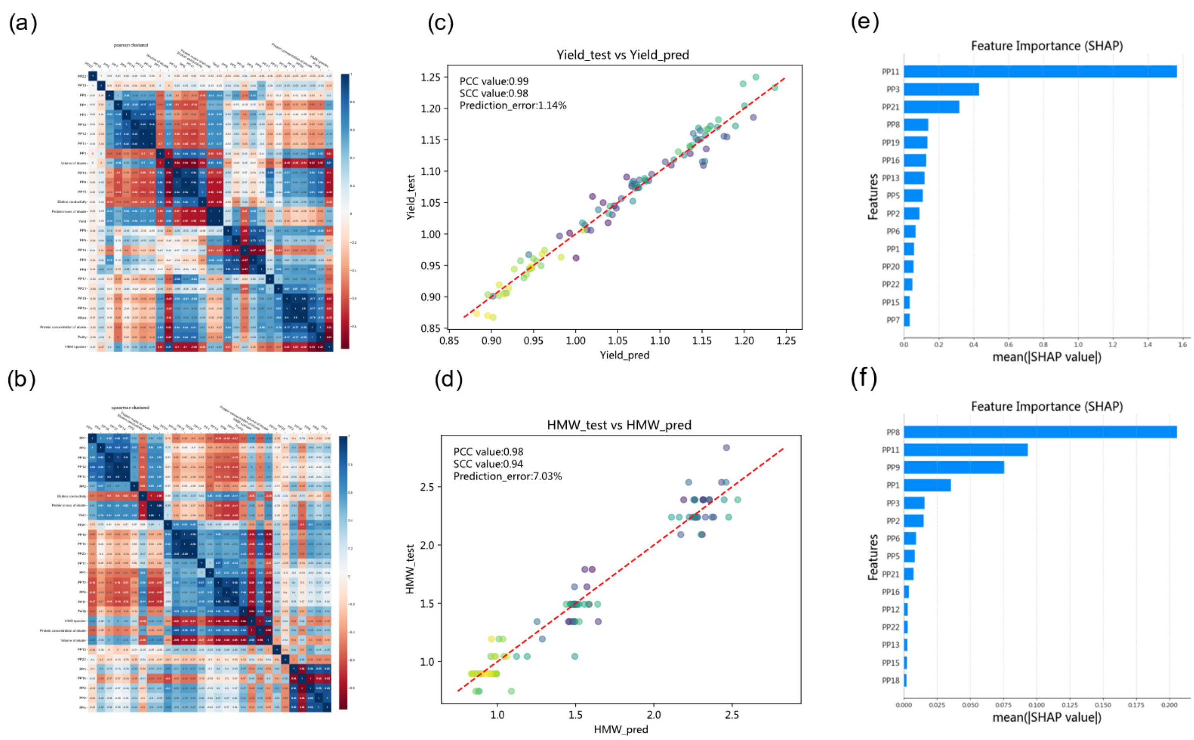

Figure 2. Crucial results of machine learning workflow. a, b, Heatmaps showing the Pearson (a) and Spearman (b) correlation matrices for four categories of process parameters. This visualization utilizes a color gradient (accessible in the online version) to indicate the strength and direction of the correlations, offering a clear overview of the interconnections among the parameters. Hierarchical clustering was applied to the Pearson/Spearman correlation matrix, with distances calculated based on absolute correlations to group both strong positive and negative correlations together. c, d, Model performance in predicting critical quality attributes (CQAs) for the test (validation) set is evaluated, where X_test represents the actual values, and X_pred denotes the model's predictions. The Pearson Correlation Coefficient (PCC), Spearman Correlation Coefficient (SCC), and Mean Absolute Percentage Error (Prediction_error) are illustrated in the figure. e, f, SHAP (SHapley Additive exPlanations) summary plots illustrating the top feature contributions to Critical Quality Attribute (CQA) predictions. Features are ranked by the magnitude of their impact on model output; factors not displayed possess negligible SHAP values. This visualization clarifies the relative significance of process parameters, enhancing model transparency and identifying key levers for process optimization.

Table 3. Selected features for machine learning.

Phase	Features/Process parameters (PPs)
Column performance	Column plates (PP1), cycles (PP2)
Sample conditioning	Diluent conductivity (PP3), diluent pH (PP4), Added NaCl Amount (PP5)
Equilibration	Conductivity (PP6), pH (PP7)
Load	Conductivity (PP8), loading density (PP9), sample loading amount (PP10)
Wash-1	Conductivity (PP11), pH (PP12)
Wash-2	Conductivity (PP13), pH (PP14), peak height[1](PP15), volume (PP16)
Elution	Conductivity (PP17), pH (PP18), eluate collection start (PP19) & end UV (PP20), temperature (PP21)

Note: Eluate collection was based on peak height in the Wash-2 phase.

chain may also cause variation in the binding and elution behaviors of PEGylated proteins on Q BB resin. However, critical raw material attributes of PEG, such as purity and molecular weight, have already been strictly controlled, resulting in low batch-to-batch variation and negligible effects. Therefore, inputs relating to the heterogeneity of the PEG chain were excluded from the ML analysis.

Incorporating these selected features into the ML models significantly enhanced the predictive accuracy for CQAs (Figure 2c,d). This feature selection process ensures that the ML algorithms remain data-driven while aligning with biochemical and process principles, thereby providing actionable insights for process optimization and quality control. These insights directly support enhancements in both process efficiency and product quality.

By integrating insights from correlation heatmaps and feature importance analysis, critical process parameters for optimization were identified. Among these, three key input process parameters – 'Load Conductivity (PP8),' 'Wash-1 Conductivity (PP11),' and 'Wash-2 Conductivity (PP13)' - had significant impact on process outcomes (Figure 2e,f). While these parameters were prioritized for exploration, numerous combinations of other critical parameters were also derived to ensure comprehensive optimization.

To explore these parameter combinations, advanced techniques such as LHS,[39] grid sampling, Bayesian optimization,[40] and genetic algorithms[41] were used, generating outcome predictions from over 10 billion parameter sets. These combinations undergo high-throughput prediction and virtual screening using a pre-trained optimal model to identify configurations that align with desired CQAs. As a representative example of this workflow, we have provided Supplementary Table S2, which details the results of an *in silico* optimization search encompassing over 2 million parameter combinations. To ensure this extensive dataset remains interpretable and accessible within standard spreadsheet constraints, we applied a hierarchical clustering analysis to aggregate the outputs into 200 representative groups, highlighting the top-performing candidates from each cluster. For transparency, the data are presented as normalized values relative to the historical baseline (defined as 1.0% or 100%). This normalization allows for a quantitative assessment of the optimization magnitude, specifically the successful decoupling of the yield/purity trade-off, while maintaining the confidentiality of proprietary information.

The scale of computational experimentation far exceeds the practicality of traditional wet lab or *in silico* mechanistic methods. Through this virtual screening process, hundreds of parameter combinations leading to improved yield/CQAs were identified within the existing proven acceptable ranges. Promising candidates are further verified through lab-scale experiments, and if results demonstrate stability and reliability, they are implemented in pilot or commercial production. This ML-driven workflow establishes a closed-loop system for efficient process optimization. However, our analysis revealed that yield improvements could not exceed 7% without compromising the percentage of HMW impurities. We are keen to explore whether higher yield increases can be achieved while maintaining or reducing HMW levels.

The analysis shown in Figure 2a,b indicates that yield and HMW removal are strongly anti-correlated within the current operating window, meaning that improvements in one normally compromise the other. To break this trade-off, conditions beyond the existing operating ranges must be explored. To accomplish this objective, rather than relying solely on ML, which is inherently limited to historical data confined to these ranges, we developed a mechanistic model that explicitly represents the fundamental principles of mass-transfer and binding phenomena. This mechanism-based approach enables exploration beyond current process boundaries, uncover new operating ranges, and achieve simultaneous improvements in both yield and HMW clearance.

In silico optimization using mechanistic model

To refine the key process parameters identified by AI/ML and enhance process outcomes, mechanistic modeling was integrated to provide a quantitative and systematic approach for understanding and precisely predicting process performances. The mechanistic model for AEX combined the EDM for mass transfer and the SMA model for adsorption. The mechanistic model was calibrated using four gradient elution experiments covering a loading density of 1.0 to 3.8 g/L and gradient elution lengths of 15 and 20 CV. The SMA model parameters were estimated with normalized root mean square errors of less than 0.031 for all calibration experiments (Figure 3a), demonstrating excellent model fidelity. The verification experiments further confirmed the model's robust extrapolation and predictive capabilities (Figure 3b).

For process optimization with the mechanistic model, six key operating parameters (salt concentrations and column volumes of the Load, Wash-1, and Wash-2 stages) were focused, which were guided by ML-identified critical variables. A stepwise optimization strategy using genetic algorithms was conducted to balance the yield improvement with process complexity. Initial optimization of the Load stage alone was

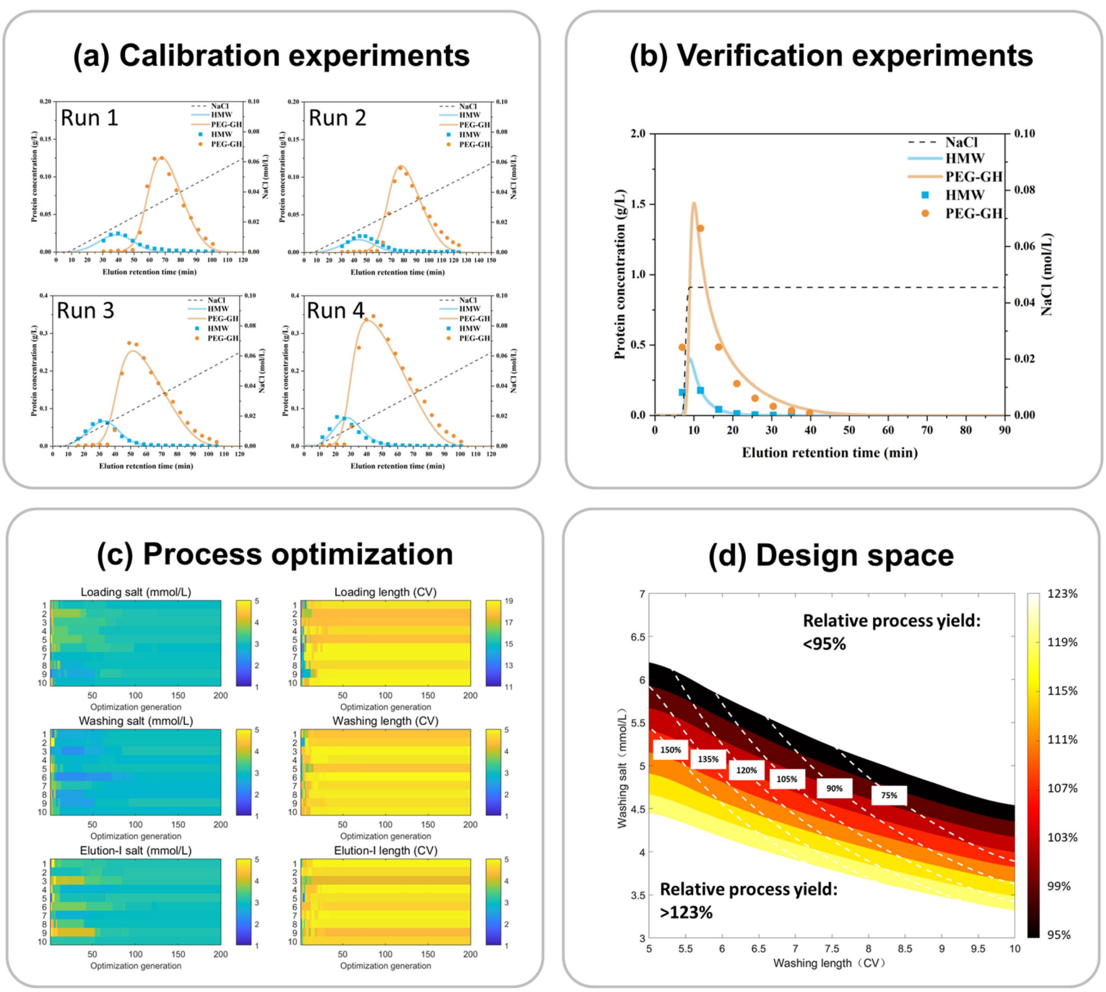

Figure 3. Workflow of mechanistic modeling. (a) Calibration experiments and mechanistic model fitting results for different linear salt gradient elution lengths and loading densities. The points represent experimental data, while the lines represent model simulations. Different colors represent various components. (b) Verification experiment and model simulation under salt stepwise elution. The points represent experimental data, and the lines represent model simulations. Different colors represent different components. (c) Ten runs of process optimization across six operating parameters. Color bars on the left represent salt concentration at the Load, Wash-1, and Wash-2 stages. Color bars on the right represent column volumes at the Load, Wash-1, and Wash-2 stages. (d) Contour map of design space. The average process yield and HMW level of the existing commercial process were normalized to one, and the other data were compared with the averages as relative values. The color bar represents relative process yield, while the white dashed lines represent the relative HMW content.

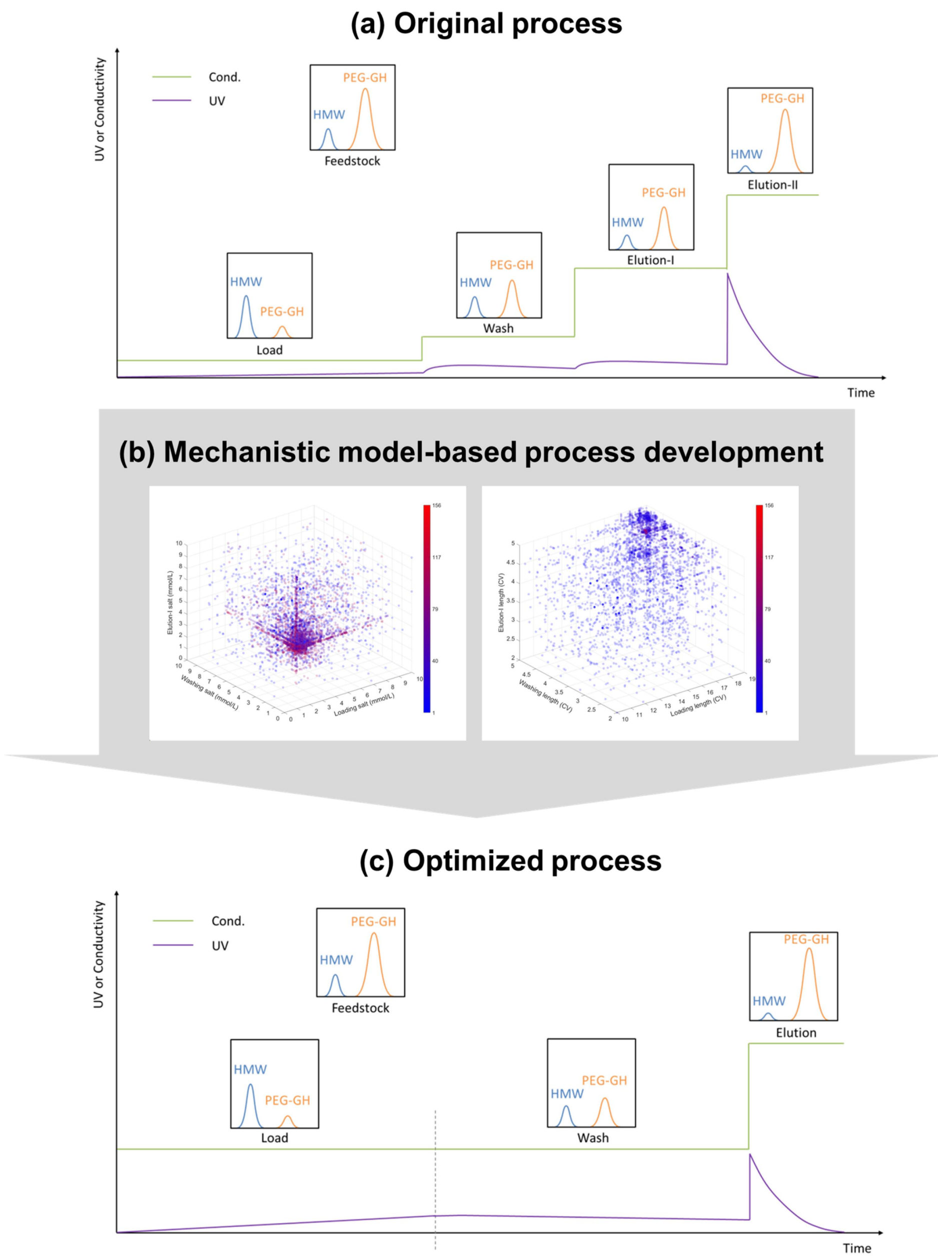

Figure 4. Process improvement by mechanistic modeling. (a) Original process diagram. The sub-figures above the conductivity curve represent the HPLC diagrams. (b) Mechanistic model-based process development. Color bars represent the optimization generations. (c) Optimized process diagram. The sub-figures above the conductivity curve represent the HPLC diagrams.

insufficient to meet the targets. Therefore, optimization was extended to include parameters from the Wash-1 and Wash-2 stages. As shown in Figure 3c, the results of 10 genetic algorithm optimization runs (totaling 46,500 *in silico* simulation) showed a high degree of consistency, with an average coefficient of variation of 3.19% across the six operating parameters. The optimization results revealed that salt concentrations of the Load, Wash-1, and Wash-2 stages converged, while column volumes approached their set limits. This led to the proposal of a process simplification: merging the Wash-1 and Wash-2 into a single wash stage, and unifying salt concentrations across these stages (Figure 4). The lab-scale verification study demonstrated a significant improvement in yield (increased by 7%) and HMW removal (decreased by 33%), indicating that the yield-purity trade-off can be overcome by implementing a hybrid ML/mechanistic modeling framework.

In addition, sensitivity analysis was performed using the mechanistic model to assess the impact of operational parameter variations. Figure 3d presents the design space for the optimized process with target constraints. The contour map shows that even minor deviations in salt concentration resulted in notable effects, with approximately 4% fluctuations in yield and 30% fluctuations in HMW level. These findings underscore the critical importance of sub-millimolar precision in salt concentration for effective process control. This aligns with the high SHAP contribution of the conductivity parameter in the ML model, extending beyond statistical correlation by quantifying the underlying physical and chemical mechanisms.

Process confirmation and robustness evaluation

To confirm the robustness of the optimized process, six verification runs were conducted: three at 46 mL column volume (lab-scale) and three at 1.8 L (pilot-scale). Additionally, DoE was used to evaluate the impact of variations in wash conductivity and volume on yield and HMW impurity content. As shown in Figure 5a,b, compared to control data from the 423 commercial manufacturing runs, the optimized process demonstrated higher yield and superior HMW removal, achieving a 6%–12% increase in yield and a 4%–33% reduction in HMW level across different scales. Meanwhile, the relative HMW content is notably

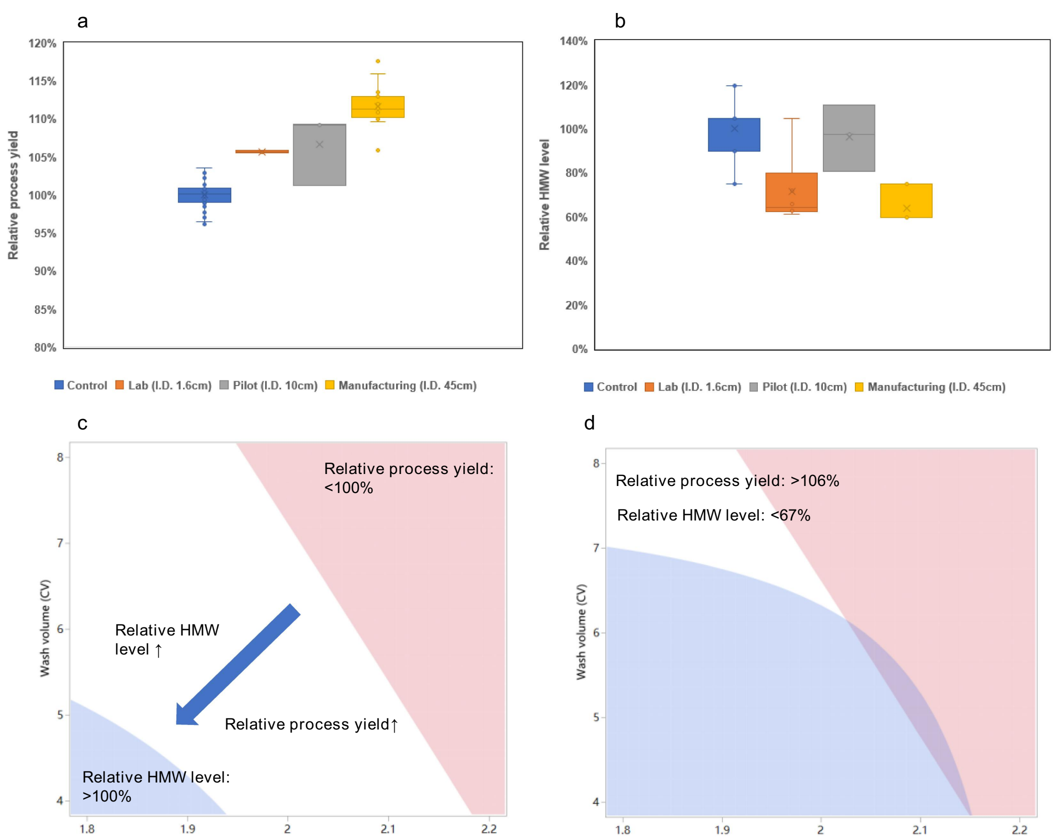

Figure 5. Process confirmation and robustness evaluation. Comparison of the optimized process with the existing commercial process in terms of (a) relative process yield and (b) relative HMW level. The control represents the existing commercial process in a chromatographic column with an internal diameter (I.D.) of 45 cm. The lab-scale (I.D. 1.6 cm), pilot-scale (I.D. 10 cm) and commercial manufacturing-scale (I.D. 45 cm) indicate the optimized process verified in columns with I. D. 1.6 cm, I.D. 10 cm and I.D. 45 cm, respectively. The average process yield and HMW level of the existing commercial process were normalized to 1, and the other data were compared with the averages as relative values. (c) Contour plot (white area) showing the operating ranges of the optimized process with comparable performance to the existing commercial process. As indicated by the big blue arrow, with the decrease of wash volume and wash conductivity, the relative process yield and relative HMW level would increase simultaneously. (d) Contour plot (white area) showing the operating ranges of the optimized process with superior performance to the existing commercial process.

higher at the pilot scale (Figure 5b). This is likely due to the relatively higher agitation, which led to the increase of HMW level since this molecule is sensitive to the shear force.

To further assess process robustness, the wash buffer conductivity range (2.0 ± 0.2 mS/cm) and wash volume range (6.0 ± 2.0 CV) were evaluated, which were carried out before PPQ. As shown in Figure 5c, when wash conductivity was adjusted within ±0.1 mS/cm (sub-millimolar variation) and wash volume within ±1.0 CV around the optimized set points, the process exhibited either higher purity at the same yield or increased yield while maintaining equivalent purity compared to the control. These results were consistent with predictions from the mechanistic evaluation (Figure 3d). A representative chromatogram of the AEX process with Q BB after optimization is shown in Supplementary Figure S1, which was well simulated by the mechanistic model. Furthermore, the process could be further modulated to achieve both higher yield and lower HMW level, as shown in Figure 5d. These findings highlight the importance of mechanistic models for process optimization within narrow operating ranges while ensuring higher yield and purity.

Although these operational ranges may appear narrow in the context of commercial manufacturing, historical data suggest that these parameters can be effectively controlled. The optimized process was verified through six full-scale PPQ runs. As shown in Figure 5a,b, the optimized process resulted in a 12% increase in yield and a 33% reduction in HMW content compared to the control process. Based on these promising results, the process changes were successfully implemented. Importantly, the yield improvement is expected to generate annual cost savings of several million dollars, underscoring the substantial economic benefit of this optimization.

Discussion

This study introduces a novel hybrid framework that integrates data-driven ML with physics-based mechanistic modeling to optimize biopharmaceutical downstream process. This approach is particularly effective for processes with large volumes of manufacturing data, where the challenge of improving yield while simultaneously reducing impurities level remains substantial. The hybrid framework effectively addresses the yield-purity trade-off and has successfully transitioned from lab-scale studies to commercial-scale manufacturing, making a groundbreaking integration of ML and mechanistic modeling. The success of this framework stems from innovative methodologies that leverage the complementary strengths of both ML and mechanistic modeling.

A key distinguishing feature is that our ML algorithms utilized commercial manufacturing data, setting it apart from traditional methods such as DoE or high-throughput screening experiments. Commercial datasets provide a more representative view of true process variability compared to laboratory-generated data. By analyzing more than 400 commercial batches, our ML methodology successfully identified three critical process parameters out of over 20 potential candidates. This data-driven approach effectively uncovers complex, non-linear relationships that are often missed by conventional statistical approaches, mitigating the subjectivity of traditional risk assessment methods and reducing the need for additional experimentation. Importantly, the accuracy of ML improves with increased availability of commercial data. However, optimizing within the current parameter range using ML alone does not meet the dual objective of increasing yield while reducing HMW content simultaneously. To address this limitation, we developed a mechanistic model to explore new operating ranges beyond those defined by existing parameter range. By explicitly representing binding and mass-transfer phenomena, the mechanistic model provides a predictive, physics-based foundation for identifying new operating conditions that could not be uncovered by ML alone.

While mechanistic model can accurately predict chromatographic profiles and facilitate process optimization, achieving optimal conditions requires the exploration of a vast number of parameter combinations, which is often computationally intensive and experimentally impractical. Unlike other hybrid methodologies,[42] our framework integrates ML to enhance the mechanistic model's performance. By first identifying the most influential critical process parameters, ML streamlines the mechanistic model, enabling it to focus on the variables most relevant to process performance and regulatory risk. This integration reduces experimental workload by over 90% and significantly enhances the process understanding. Notably, this hybrid framework improved both yield and product quality of the existing commercial process without requiring major modifications, demonstrating its feasibility and scalability for industrial implementation.

The process procedure was consistent across laboratory, pilot, and commercial scales. The process change was categorized as moderate following communication with the regulatory agency, and approval was granted based on a comprehensive analytical comparability study. This study focused on pre- and post-change products, including rigorous assessment of identity, purity, potency, stability, and other CQAs. The data demonstrated high comparability, with no adverse impact on product quality and improved HMW removal. Therefore, clinical or non-clinical data were not required for this moderate change.

Despite these successes, the framework faces certain limitations. Its effectiveness is contingent on data availability, requiring a substantial dataset (e.g., >30 batches) for reliable ML predictions. Furthermore, the current mechanistic model focuses on HMW impurities, while other CQAs, such as positional isomers, remain to be integrated. Finally, the model currently assumes consistency in resin attributes (e.g., ligand density, pore size); future work should incorporate this material variability to enhance predictive accuracy and robustness.

A key limitation of the present work is its reliance on historical data for offline optimization, lacking the capability for real-time control. The implementation of real-time Process Analytical Technology (PAT) tools, such as online Raman or near-infrared spectroscopy, remains a critical future step to enable dynamic adjustments of critical process parameters based on live monitoring of CQAs[43–48]. The implementation of these in-line or on-line PAT tools will require a rigorous, lifecycle approach encompassing method development, model validation, and regulatory alignment. This includes qualifying PAT sensors (IQ/OQ/PQ), developing and validating robust chemometric models to ensure predictive accuracy, and establishing a control strategy that defines how real-time data will be used to adjust critical process parameters. Furthermore, this effort will involve engaging with evolving regulatory frameworks [49,50] for PAT and Real-Time Release Testing (RTRT), and fostering cross-sector collaboration to advance the development of adaptive manufacturing systems based on communal datasets and shared best practices.

In summary, the hybrid ML-mechanistic framework developed in this study offers distinct advantages for biomanufacturing, including efficiency through a dramatic reduction in experimental workload, scalability across equipment scales, and enhanced regulatory compliance by coupling data-driven insights with mechanistic understanding. Based on historical data from over 400 commercial lots, three critical process parameters were identified via ML, reducing the number of parameters optimized by mechanistic models. Subsequent *in silico* optimization across more than 40,000 scenarios successfully broke the yield-purity trade-off, resulting in a 12% increase in yield alongside a 33% reduction in HMW impurities. The optimized process condition was successfully verified by 6 full-scale PPQ runs (18 runs of AEX in total), consistently demonstrating scalability and process robustness. The process change was categorized as moderate, and approval was granted based on a comprehensive analytical comparability study.

Ultimately, this work lays the foundation for the future of smart biomanufacturing and Industry 4.0, where AI-driven process control and automated, adaptive systems will drive the next generation of biopharmaceutical production, ensuring higher yields, lower costs, and better product quality. Moreover, future research should focus on digital twins integrating real-time PAT and RTRT to enable dynamic process control for intelligent biomanufacturing.

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Author contributions

Rong-Zhu Wang: conceptualization, formal analysis, methodology, visualization, writing – original draft, project administration. **Yu-Xin Liao**: data curation, formal analysis, methodology, software, visualization, writing – original draft. **Teng-Long Wang**: data curation, formal analysis, methodology, software, visualization, writing – original draft. **Xiao-Yang Jing**: data curation, formal analysis, methodology, software, visualization, writing – original draft. **Guang Yang**: data curation, formal analysis, methodology, software, visualization. **Hao Wang**: investigation. **Feng Xiao**: methodology. **Yi Zong**: methodology. **Yan Yang**: methodology, supervision and writing – review & editing. **Ye Yuan**: methodology. **Shan-Jing Yao**: writing – review & editing. **Xuan Xian**: supervision. **Lei Jin**: supervision. **Dong-Qiang Lin**: supervision and writing – review & editing. **Yun (Kenneth) Kang**: supervision and writing – review & editing.

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List of abbreviation

AEX	anion-exchange chromatography
AI	artificial intelligence
ANN	artificial neural network
CHO	Chinese hamster ovary
CPP	critical process parameter
CQA	critical quality attribute
CV	column volume
DoE	design of experiments
EDM	equilibrium dispersive model
FDA	Food and Drug Administration
HMW	high molecular weight
HPLC	high performance liquid chromatography
ICH	International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use
IoT	Internet of things
I.D.	internal diameter
KPP	key process parameter
LHS	Latin hypercube sampling
ML	machine learning
PAT	process analytical technology
PP	process parameter
PPQ	process performance qualification
PEG	polyethylene glycol
QbD	Quality by Design
R^2	coefficient of determination
RTRT	real-time release testing
SEC	size exclusion chromatography
SHAP	SHapley Additive exPlanations
SMA	steric mass action
UV	ultraviolet

Data availability statement

The data are not publicly available due to some information that could compromise the privacy of the commercial process.

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